

XML Update Utility - Full Development Roadmap

Developer Planning Guide for Java 8 + Eclipse + Maven

Project Objective

Update specific fields inside claimMember sections of XML files using Excel as the source of truth while preserving the original XML formatting, namespace ordering, XML declaration, whitespace, indentation, and empty-tag representation.

Business Scope - Current

Fields to update: submittedMemberID, lastName, firstName, middleName.

Design Decision

DOM/XPath/Transformer approach is intentionally avoided because it modifies XML declaration, namespace order, and empty-tag formatting. The solution will use a controlled text-processing approach to preserve file integrity.

Target Processing Flow

- 1 Read Excel once
- 2 Load Excel records into HashMap
- 3 Read XML as text
- 4 Locate claimMember block
- 5 Extract submittedMemberID
- 6 Lookup Excel data
- 7 Update target values only
- 8 Create middleName if missing
- 9 Save XML to target folder
- 10 Generate processing logs

Architecture

MainApplication -> ExcelReader -> MemberData -> XmlFileProcessor -> ClaimMemberUpdater -> XmlReplacementUtil -> Logger

Project Structure

com.ics.claims.main, model, service, util, config

Development Phases

- 1 Phase 1: Environment Setup (Java 8, Maven, Eclipse, project structure).
- 2 Phase 2: Model Layer (MemberData POJO).
- 3 Phase 3: ExcelReader using Apache POI and HashMap lookup.
- 4 Phase 4: XML File Reader and output folder handling.
- 5 Phase 5: claimMember block discovery and submittedMemberID extraction.
- 6 Phase 6: Controlled replacement engine for submittedMemberID, firstName, lastName, middleName.
- 7 Phase 7: Missing middleName node creation logic.
- 8 Phase 8: Exception handling and logging.
- 9 Phase 9: Batch processing for multiple XML files.
- 10 Phase 10: Unit testing and validation.

Class Responsibilities

- 1 MainApplication: Entry point and orchestration.
- 2 MemberData: Represents one Excel row.
- 3 ExcelReader: Reads Excel and builds lookup map.
- 4 XmlFileProcessor: Reads and saves XML files.
- 5 ClaimMemberUpdater: Finds claimMember sections and performs updates.
- 6 XmlReplacementUtil: Safe tag replacement helper methods.
- 7 ProcessingLogger: Success, warning, and error logging.

Exception Matrix

No Match -> Skip and log. Duplicate Excel IDs -> Error and log. Missing middleName -> Create node. Invalid XML content -> Log and skip file.

Future Enhancements

DOB, Gender, Address, Provider Information, Payer Information, Diagnosis Information, Config-driven field mapping, UI and scheduler support.

Definition of Success

Output XML structure must remain identical to input XML except for approved business field updates.